AI AISSISTED CODING LAB – 1

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Batch:24

Lab Assignment -1: Comparing Three Ai Codes

Promt: Write a python code of prime numbers from 1 to 20

Github Copilot:

```
primes = []
for num in range(2, 21):
    is_prime = True
    for i in range(2, int(num ** 0.5) + 1):
        if num % i == 0:
            is_prime = False
            break
    if is_prime:
        primes.append(num)

print("Prime numbers from 1 to 20:")
print(primes)
```

#output

Prime numbers from 1 to 20:

[2, 3, 5, 7, 11, 13, 17, 19]

Goggle Collab

#code

```
def is_prime(num):
    if num <= 1:
        return False
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            return False
    return True

prime_numbers = [num for num in range(1, 21) if is_prime(num)]
print(prime_numbers)
```

#output

[2, 3, 5, 7, 11, 13, 17, 19]

Claude

#code

```
def is_prime(n):
    if n < 2:
        return False
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True

primes = [n for n in range(1, 21) if is_prime(n)]
print(primes)
```

#Output

[2, 3, 5, 7, 11, 13, 17, 19]

Conclusion:

The three AI-generated Python codes successfully identify and display the prime numbers from 1 to 20. GitHub Copilot, Google Colab, and Claude used slightly different coding approaches, but all produced the same correct output: [2, 3, 5, 7, 11, 13, 17, 19]. This comparison shows that AI tools can generate accurate

and efficient Python programs, while also demonstrating that different AI tools may use different programming styles to solve the same problem.